Syllabus: Surrogate Modeling for Process Prediction and Optimal Decision-Making

Build validated process predictors, learn optimization formulations and model structures, then select and revalidate operating conditions.

Course design and notes: Sunwoo Kim · Graduate entry

Course purpose

Define inputs and outputs, fit a surrogate to process-response data, and validate its predictions under independent conditions. Then assess physical consistency, represent the learned function in an optimization problem, and select operating conditions for revalidation with the reference process model.

The course covers both formulating an already-trained model for optimization and designing a model structure for that use. Prediction error, physical consistency, formulation agreement, and decision quality are evaluated separately.

Sunwoo Kim designed this eight-week, self-paced course for chemical engineers entering graduate study. Original explanations, worked readings, and runnable examples accompany verified research sources.

Audience and prerequisites

Basic Python, derivatives, matrix operations, linear algebra, and material balances are expected. LP/MILP, convexity, and KKT conditions are introduced when needed, so prior neural-network or mathematical-programming research experience is not required.

Learning outcomes

By the end of the course, learners should be able to:

1. Define a process prediction task: inputs, outputs, fixed context, states, units, data splits, and valid domain.
2. Fit steady-state and dynamic surrogates, and evaluate component errors and trajectory errors using independent data and prescribed future inputs.
3. Compare unconstrained, soft-physics, and hard linear-equality models, separating prediction accuracy from physical consistency.
4. Define operating variables, an objective, and constraints; formulate a frozen ReLU model as a valid MILP and check agreement with its forward pass.
5. Select model structure and formulation for the intended use, including the conditions under which ReLU ICNN/PICNN epigraphs permit an equivalent LP problem.
6. Solve a surrogate-based operating problem and report reference-model feasibility, economic performance, solver termination, gaps, and time separately.

Eight-week sequence

Week	Topic and scope	Required output
1	**Process Prediction and Surrogate Modeling Basics.** Inputs, outputs, fixed context, units, operating domain, reference data, a regression baseline, and independent prediction checks.	A steady-state CSTR data contract, fitted regression baseline, and held-out prediction report.
2	**Learning Multicomponent Steady-State Responses with MLPs.** Three-component balances, scaling, MLP forward/backpropagation, training, component errors, and concentration consistency.	A steady-state MLP evaluated in physical units, with a hand-checked gradient update.
3	**Predicting Dynamic Responses and Rollout.** State-transition MLPs, initial states, future operating inputs, one-step prediction, and full-trajectory rollout. RNN/LSTM and encoder–decoder extensions are optional.	A comparison of one-step and trajectory errors under prescribed future inputs.
4	**Learning with Physical Constraints.** Residual losses, automatic differentiation, hard linear-equality projection, KKT derivation, rank assumptions, and remaining violations.	An MLP/soft-physics/hard-equality comparison separating accuracy from physical consistency.
5	**From a Predictor to an Optimization Formulation.** Operating variables versus fixed context, objectives and constraints, frozen ReLU graphs, binary variables, Big-M, valid bounds, and scaling.	A hand-built MILP representation checked against the frozen network forward pass.
6	**Solve Time and Results in Neural-Network Optimization.** Bound tightening, scaling, OMLT, LP relaxations, HiGHS termination, gaps, computation time, and reference-process checks.	A comparison of formulations for the same frozen network, reporting bounds, binary counts, termination, gaps, timing, and forward checks.
7	**Designing Model Structure for Optimization.** ICNN/PICNN, convexity conditions, ReLU/linear epigraphs, fixed context, inequality direction, and equivalence counterexamples.	An ICNN and valid LP on a known convex example, with explicit applicability conditions.
8	**Selecting and Revalidating Operating Conditions.** Decision-point errors, extrapolation, physical feasibility, economic performance, reference-model checks, and targeted additional data.	A reproducible prediction-to-optimization report separating errors, violations, decision quality, and computation.

Core sources

- Stephen Boyd and Lieven Vandenberghe, ***Convex Optimization*** (Cambridge University Press, 2004).
- Stephen Boyd, **EE364a: Convex Optimization I — Lecture Slides**, Stanford University.
- **cog-imperial/OMLT**: Optimization and Machine Learning Toolkit.
- **li-group/KKThPINN**: KKT-hPINN code and datasets.